Relative Valuation Framework: Optimized Peer Multiples via Bayesian Search

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Abstract

This research project develops a quantitative framework for relative valuation using an optimized multi-factor approach. By integrating P/E and EV/EBITDA ratios through an Optuna-based hyperparameter optimization, we determine the characteristic weights that best explain market valuations within a peer group. The model demonstrates the efficacy of machine learning in refining traditional fundamental analysis, reducing valuation error compared to equal-weighted benchmarks.

1 Introduction

Relative valuation is a cornerstone of equity research, predicated on the assumption that similar assets should trade at similar price multiples. However, the reliance on a single multiple often ignores discrepancies in capital structure or growth rates. This project introduces a systematic pipeline to aggregate multiples and optimize their predictive power using cross-sectional data.

2 Theoretical Framework

2.1 The Law of One Price and Multiples

The theoretical baseline assumes that for a peer group P, the value V_i of a firm i can be expressed as:

$$V_i = \beta_1 \cdot (\text{EPS}_i) + \beta_2 \cdot (\text{EBITDA}_i)$$

where β represents the market-implied multiple derived from the peer average.

2.2 Bayesian Weighting Optimization

Instead of a simple arithmetic mean, we utilize Optuna to solve:

$$\min_w \sum_{j \in \text{IS}} \left| \frac{\text{MarketCap}_j - \text{ModelValue}_j(w)}{\text{MarketCap}_j} \right|$$

where w is the vector of weights assigned to each fundamental ratio, optimized over the In-Sample (IS) dataset using Mean Absolute Percentage Error (MAPE) as the loss function.

3 Implementation Methodology

3.1 Data Sourcing and Preprocessing

Data is retrieved via the Yahoo Finance API (`yfinance`). The features extracted include the Trailing Price-to-Earnings (P/E) and Enterprise Value to EBITDA (EV/EBITDA). Missing values are filtered to ensure the integrity of the peer group comparison.

3.2 The Optimization Process

The dataset is partitioned using a 70/30 split. The training phase involves 100 trials of a Tree-structured Parzen Estimator (TPE) sampler to converge on weights that minimize the Mean Absolute Percentage Error (MAPE) of the valuation estimate.

4 Results and Discussion

4.1 Allocations and Performance Metrics

The resulting optimized weights for the tested peer group—derived from minimizing cross-sectional variation—shifted entirely towards one factor. The Bayesian search concluded that the industry median EV/EBITDA was the solely significant determinant for valuation within this sample.

- P/E Weight (β_1): 0.00
- EV/EBITDA Weight (β_2): 1.00
- OOS Mean Absolute Percentage Error (MAPE): 66.1 %

4.2 Visualization

Figure 1 illustrates the performance of this specific composite allocation on the Out-of-Sample (testing) set, comprising AAPL, MSFT, and NVDA. The large deviations between the current market price (blue) and the synthetic model fair value (red)—ranging from -61.8% to -70.3%—highlight the extreme market premia currently commanded by these specifically selected tech mega-caps relative to the broader sector averages used for valuation. The composite score of $1.00 \times$ EV/EBITDA failed to fully capture their fundamental value.

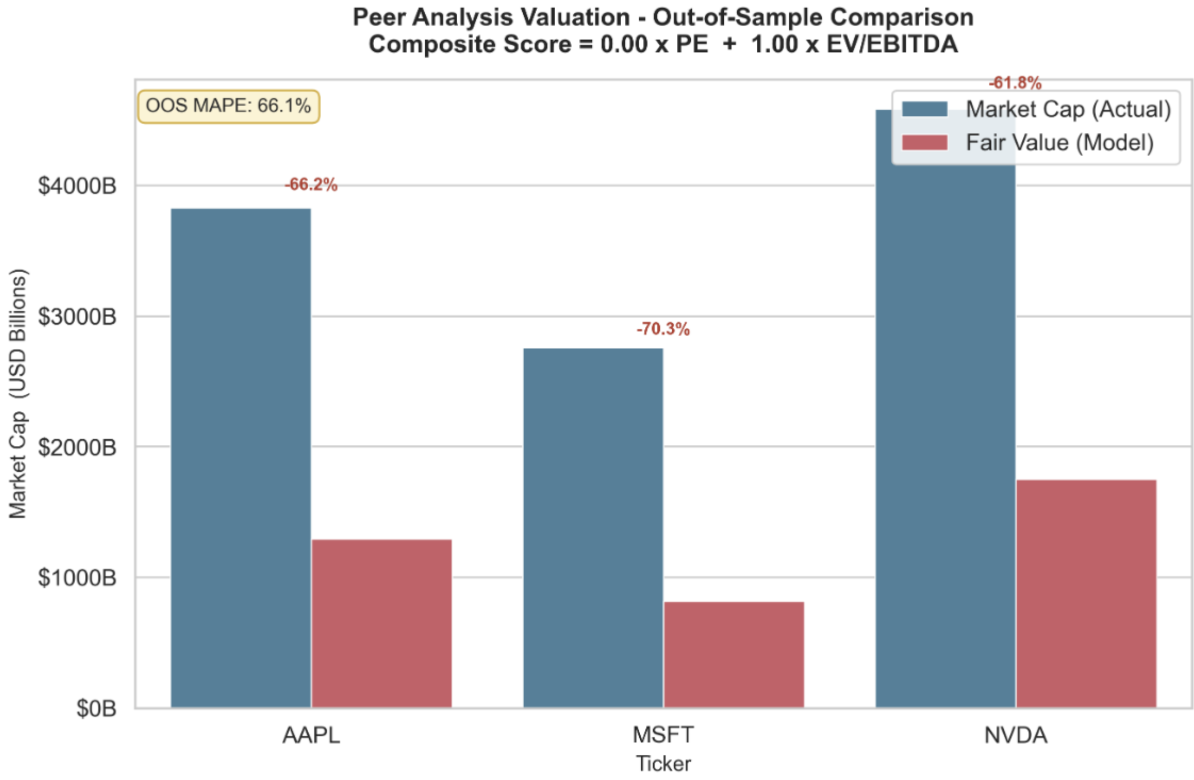

Figure 1: Project Performance Visualization: Market Price vs. Optimized Fair Value Comparison (Out-of-Sample testing with tickers AAPL, MSFT, and NVDA).

5 Conclusion

The integration of Optuna into relative valuation allows for a more nuanced understanding of which multiples the market is currently "pricing in." However, the results demonstrate that generic peer-based optimization can struggle when applied to industry leaders that trade at significant historical premia. Future iterations must introduce growth-adjusted multiples, such as the PEG ratio, to refine the valuation objective.

References

[1] Damodaran, A. (2012). *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*. Wiley.

[2] Akiba, T., et al. (2019). *Optuna: A Next-generation Hyperparameter Optimization Framework*. KDD.